Problem 18.12

A trust has been established such that RJ will receive a perpetuity of \$1,000 a year with the first payment at the end of 5 years. Calculate the present value of the perpetuity at a discount rate of $d = 8\%$.

Solution

The effective annual discount rate is

$$d = 0.08.$$

The corresponding effective annual interest rate is found from

$$d = \frac{i}{1+i}.$$

Thus,

$$0.08 = \frac{i}{1+i}.$$

Solving,

$$i = \frac{0.08}{1-0.08} = \frac{0.08}{0.92} \approx 0.0869565.$$

The first perpetuity payment is at the end of year 5. A perpetuity-immediate with first payment at time 5 is valued one period before the first payment, at time 4. Its value at time 4 is

$$\frac{1000}{i}.$$

Discounting this value back four years gives

$$PV = \frac{1000}{i} v^4.$$

Since $v = 1 - d = 0.92$,

$$\begin{aligned} PV &= \frac{1000}{0.0869565}(0.92)^4 \\ &\approx 8238.5190. \end{aligned}$$

Equivalently,

$$PV = 11500(0.92)^4 \approx 8238.5190.$$

Hence, the present value of the perpetuity is

$$\boxed{\$8{,}238.52}.$$